

RSCTF TECHNICAL PRACTICE PAPER

RSCTF KING OF THE HILL HANDBOOK: BOOT2ROOT AND LEADERBOARD FORMATS

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Scoring policy: one constant formula per KotH format; no scoring versions

ABSTRACT

RSCTF defines two King of the Hill (KotH) formats with different competitive meanings. **Boot2Root KotH** is an exclusive-control contest over one shared machine: teams acquire the hill, retain control, and accept responsibility for service health. **Leaderboard KotH** is a concurrent application or protocol contest: every eligible team can play in the same tick, challenge-native outcomes are normalized by RSCTF, and consistently remaining first earns a bounded bonus. Boot2Root uses the constant score $100R(0.25A + 0.55C + 0.20\sqrt{AC})$. Leaderboard uses a per-tick harmonic activity/objective performance rate, exact tied-leader credit, and adjacent-tick continuity. Failed hacking attempts are not negative points; authenticity, replay protection, exact runtime identity, independent checking, and challenge-level proof design determine whether evidence is admitted. Boot2Root capabilities rotate with pristine crown cycles; a Leaderboard capability is one opaque event token and changes only through explicit security rotation. Both formats use scoped capabilities, pristine crown-cycle replacement, bounded hill aggregation, and finalized epoch settlement. This paper is the canonical deployed scoring and operations contract.

Keywords: King of the Hill; Boot2Root; Leaderboard KotH; normalization; sustained lead; anti-cheat; crown cycle; RSCTF

Document status: current implementation contract for the repository revision containing this paper. Public names are Boot2Root and Leaderboard; the stable internal claim-source identifiers remain **Marker** and **Api**. Those identifiers are not selectable scoring versions. This is a technical-practice report, not a claim of peer review or empirical validation.

Two meanings of “hold the hill”

Exclusive machine control and concurrent leaderboard dominance need different evidence

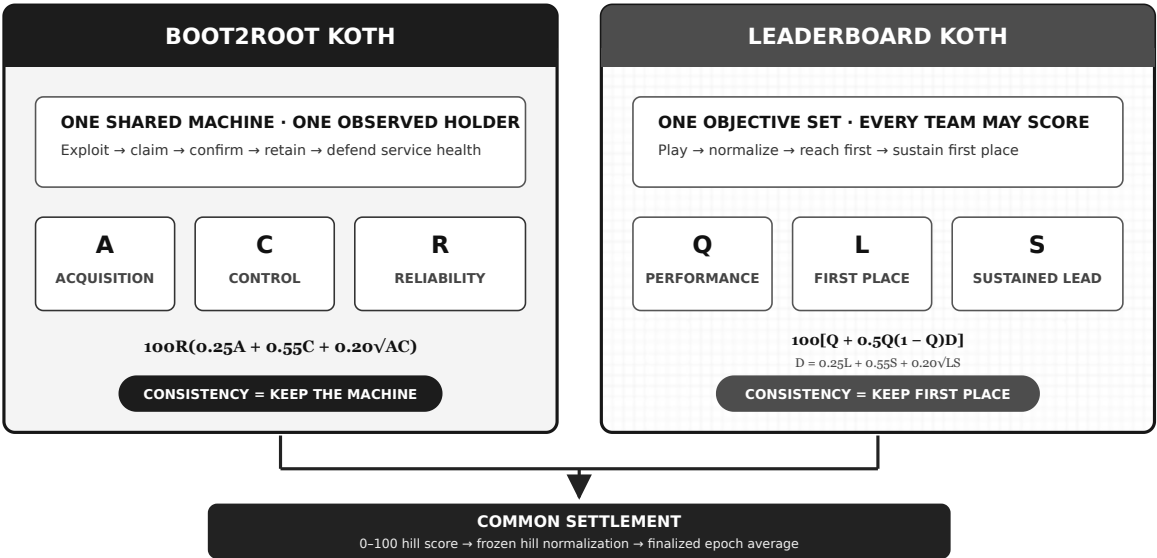


Figure 1. Boot2Root measures exclusive machine control. Leaderboard measures normalized concurrent performance and sustained first place.

START HERE: KOTH IN 90 SECONDS

Choose the hill's format

Boot2Root means one shared machine and one holder:

1. exploit or administer the hill;
2. place the current per-hill capability in `/koth/king` ;

3. remain healthy through confirmation; and
4. retain control until takeover or the pristine reset.

Leaderboard means every eligible team may score concurrently:

1. use the challenge's player-facing mechanic;
2. complete meaningful, verified work;
3. optimize every published objective; and
4. sustain first place across adjacent ticks.

Players never call the organizer's signed referee endpoint.

The two constant formulas

```

BOOT2ROOT
Core = 0.25A + 0.55C + 0.20 * sqrt(A * C)
Local = 100 * R * Core

LEADERBOARD
Q_t = 0 if E_t=0 or P_t=0; otherwise 1/(0.35/E_t + 0.65/P_t)
D = 0.25L + 0.55S + 0.20 * sqrt(L * S)
Local = 100 * [Q + 0.50 * Q * (1-Q) * D]
```

The first formula uses acquisition **A**, control **C**, and reliability **R**. The second uses tick activity **E_t**, normalized objective performance **P_t**, mean tick performance **Q**, lead coverage **L**, and sustained-lead continuity **S**.

Table 1. Fixed public formats and internal claim-source identifiers.

Public format	Internal source	Concurrent scorers	Evidence
Boot2Root	Marker	at most one observed controller	acquisition, control, reliability
Leaderboard	Api	every eligible team	activity, objectives, sustained lead

An API credential configured before official scoring selects internal source **Api**; otherwise the hill uses **Marker**. The source cannot change after scoring begins. “Leaderboard” describes the competition, not its transport: the player surface may be HTTP, a binary protocol, a simulator, or a game server.

1.2 Shared invariants

Both formats preserve these requirements:

1. **Constant policy.** Each format has one formula and fixed coefficients.
2. **Bounded result.** A hill-epoch and aggregate epoch remain in **[0, 100]**.
3. **Current identity.** Evidence belongs to the exact game, hill, round, cycle, reset attempt, target, and container. Boot2Root additionally binds the cycle capability; Leaderboard binds the current event-token generation.
4. **Independent health.** A functional checker, not the scoring feed, decides whether the shared application works.
5. **No stale carry.** A missing team in a valid Leaderboard snapshot becomes zero; a missing field snapshot is void. Neither repeats old evidence.

What resets and what settles

Several ticks form a **crown cycle**. RSCTF then pauses the hill, destroys the old container, and creates one pristine replacement from the frozen image. Boot2Root receives new cycle capabilities. Leaderboard retains each team's event capability, while transient challenge sessions, patches, implants, and referee snapshots stop working. Readiness must pass before scoring resumes.

Several cycles form an **epoch**. **Projected** includes unfinished evidence; **Settled** includes finalized epochs and determines official rank.

1. SCOPE AND DESIGN REQUIREMENTS

1.1 Why the formats are separate

A Boot2Root hill asks who controls one shared target. Its natural evidence is exclusive identity, duration, acquisition, and service responsibility. A concurrent application may instead expose puzzles, transactions, simulations, or protocol objectives that many teams can complete at once. Treating the second as a one-holder marker creates last-write races; treating the first as an external point feed transfers scoring authority away from the platform.

RSCTF therefore freezes one competitive meaning per hill:

6. **Infrastructure neutrality.** Reset, readiness, incomplete issuance, and attributable platform failures do not penalize participants.
7. **Auditable settlement.** Immutable tick evidence precedes final rollups, and retries cannot create duplicate credit.
8. **Frozen bounded state.** Rosters, hills, cadence, images, weights, objective identities, request sizes, replay memory, and retained evidence are bounded.

1.3 Relation to other systems

Bock, Hughey, and Levin describe an instructional KoH in which taking a shared machine also creates responsibility for critical services [2]. Boot2Root follows that exclusive-control interpretation while adding periodic pristine replacement and fixed epoch settlement.

CTFd documents a KoH agent that reports a current identifier and awards a configured reward per check [3]. RSCTF additionally records provisional control, confirmation, acquisition, duration, and reliability.

rCTF documents an HMAC-signed dynamic-score endpoint accepting absolute **userId/points** setters [5]. RSCTF retains exact-body signing and replay resistance, but

Leaderboard never accepts external points or team IDs. The platform normalizes bounded evidence, writes omitted-team zeros, detects leaders, and computes every score itself.

FAUST CTF is an Attack & Defense system rather than a shared hill, but its repeated rounds and service checks provide an operational comparison [4]. Its topology and formula are not RSCTF KoH.

2. COMPETITION PROTOCOL

2.1 Boot2Root protocol

RSCTF runs one managed shared container per Boot2Root hill. Every eligible team receives a high-entropy bearer capability scoped to the exact hill and reset attempt. A team claims by writing it to:

```
/koth/king
```

The checker reads the marker immediately before and after a functional probe. A stable eligible capability creates control and responsibility evidence. The first healthy observation is provisional; the configured consecutive healthy streak confirms acquisition. A different capability begins a new claim.

The previous cycle's team, or tied teams with the most confirmed healthy control, normally receives the configured opening cooldown. If blocking all tied leaders would leave no challenger, RSCTF omits the block. A blocked tick is removed from that team's personal opportunity denominator.

2.2 Leaderboard protocol

Each team receives one opaque, high-entropy capability for the game and Leaderboard hill. The player pastes only that value into the challenge. The challenge scopes it to its configured game and hill, exchanges it with RSCTF, and receives the authoritative team name plus a SHA-256 pseudonym. No local crew ID or crew name is accepted in event mode. The challenge discards the raw token and records only the lowercase pseudonym beside independently verified events. A trusted referee runs outside the player-controlled runtime.

For every active scoring round, the referee fetches a context bound to the exact runtime, objective schema, and eligible capability hashes. It filters its untrusted feed against that set and submits:

- one activity **earned/possible** ratio;
- one to sixteen objective **earned/possible** ratios; and
- an ordered **objectiveIds** list defining those positions.

The signed body contains **tokenHash**, not the bearer token. It contains no team ID, floating-point score, integrity multiplier, or point value. RSCTF resolves hashes against current capabilities and reports submitted and recognized row counts.

The capability survives pristine crown-cycle replacement. A player may explicitly rotate it after suspected exposure. Rotation immediately removes the old digest from the eligible set and clears only that team's current unsettled row; every other team's evidence and already settled epoch evidence remain immutable. The player must reconnect with the replacement token.

The first accepted snapshot freezes the exact objective IDs and order. The schema survives credential rotation. A later reorder, rename, insertion, or deletion is rejected even when the component count stays the same.

The checker allows a bounded six-second arrival window, reads the complete snapshot, performs an independent probe, and reads it again. Only a byte-equivalent current snapshot plus a healthy verdict produces one dense result row per frozen eligible team. Omitted teams receive zero. A changing snapshot or unhealthy shared application voids the field-wide tick.

Leaderboard does not elect a holder, create a provisional crown, or apply champion-cooldown scoring. Multiple teams can score in one tick.

2.3 Crown-cycle lifecycle

Both formats use the same durable sequence:

```
finalize → pause → audit/snapshot → destroy  
→ create → issue → readiness → resume
```

RSCTF stores every transition phase, prevents a second replica from owning the same operation, launches one replacement from the official image, rotates the Boot2Root window, preserves Leaderboard event tokens, and resumes only after readiness succeeds. A retry continues the stored phase instead of starting a parallel replacement. Reset and readiness time create no scoring opportunity.

From observed outcomes to a bounded KoH score

The scorer uses durable platform evidence; teams do not declare captures, patches, or intent

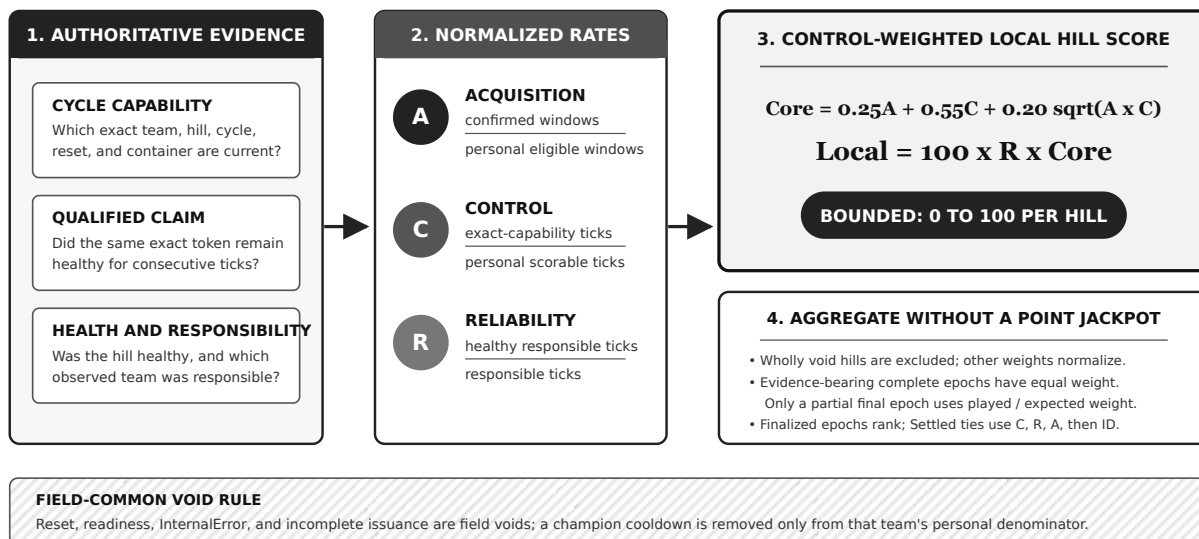


Figure 2. The checker admits current evidence; only immutable tick rows enter epoch scoring.

2.4 Exact attribution and voids

At most one immutable checker result exists per game, hill, and scoring round. Scorable evidence must match the frozen hill, active cycle, reset attempt, container, deadline, capability window, and—for Leaderboard—the objective schema.

InternalError, incomplete capability issuance, and uncertain runtime inspection are platform voids. For Boot2Root, a responsible holder may receive the last authoritative availability verdict before recovery. For Leaderboard, a shared failure is field-wide void because it cannot be fairly assigned to one participant.

3. BOOT2ROOT SCORING

For team **i**, hill **h**, and epoch **e**, define:

Table 2. Boot2Root evidence counters.

Symbol	Evidence
x	crown-cycle windows in which the team confirmed acquisition
y	windows in which the team had an eligible opportunity
u	scorable ticks controlled by the exact capability
s	personally eligible ticks after void/cooldown removal
b	responsible ticks with checker verdict OK
d	ticks for which the team was responsible

The three rates are:

$$A = x / y, \quad C = u / s, \quad R = b / d.$$

An empty denominator returns zero. The acquisition/control core is:

$$B^M = 0.25A + 0.55C + 0.20\sqrt{AC}.$$

The local hill score is:

$$H^M = 100RB^M.$$

Control has the largest direct coefficient. The geometric term rewards a team that both takes and retains the hill; reliability constrains the entire score.

3.1 Worked Boot2Root example

Suppose a team confirms one of two eligible windows, controls three of eight eligible ticks, and stays healthy in four of five responsible ticks:

$$A = 0.5, \quad C = 0.375, \quad R = 0.8.$$

$$B^M = 0.25(0.5) + 0.55(0.375) + 0.20\sqrt{0.5(0.375)}.$$

Thus **$B^M = 0.41785$** and the local score is **33.43**. Improving control raises the main term and the acquisition/control balance term; breaking the service reduces every earned control point through **R**.

4. LEADERBOARD SCORING

4.1 Independent normalization

For team i , hill h , tick t , activity evidence is $\text{activityEarned}/\text{activityPossible}$. Objective j is $\text{objectiveEarned}/\text{objectivePossible}$. Every budget is an integer satisfying:

$$0 \leq \text{earned} \leq \text{possible} \leq 10^{12}.$$

RSCTF calculates:

$$E_t = \frac{a_t^+}{a_t^*},$$

$$p_{tj} = \frac{o_{tj}^+}{o_{tj}^*}, \quad P_t = \frac{1}{m} \sum_{j=1}^m p_{tj}.$$

Each native objective is first converted to a fixed integer normalization scale. A $9/10$ result and a $900/1000$ result both contribute 0.9 ; a larger unit cannot become an accidental weight. Every objective has equal influence unless the challenge publishes a single combined objective by design.

4.2 Per-tick performance

If either required channel is zero, then $Q_t = 0$. Otherwise:

$$Q_t = \frac{1}{0.35 / E_t + 0.65 / P_t}.$$

This weighted harmonic mean makes both actual play and objective performance necessary. The objective channel has greater influence, while either weak channel remains a bottleneck. RSCTF stores Q_t before epoch aggregation, so activity in one tick cannot combine with performance in another.

Failed hacking attempts are not subtracted. Invalid, forged, replayed, stale, or wrongly scoped evidence is rejected or excluded as telemetry. The scored mechanic must still require meaningful verified work; Section 7 defines that challenge-design obligation.

4.3 Exact lead credit

For an epoch with T field-scorable ticks:

$$Q = \frac{1}{T} \sum_{t=1}^T Q_t.$$

A tick is competitive only when at least two teams have positive Q_t . If k teams share the exact highest positive value, each receives $L_t = 1/k$. Every other team receives zero. This avoids awarding a solo-field bonus and avoids a team-ID tie-break.

Lead coverage is:

$$L = \frac{1}{T} \sum_{t=1}^T l_t.$$

For $T < 2$, sustained lead S is zero. Otherwise:

$$S = \frac{1}{T-1} \sum_{t=2}^T \min(l_{t-1}, l_t).$$

The minimum preserves fractional tied-leader credit. L rewards reaching first; S distinguishes consecutive leadership from scattered peaks.

4.4 Bounded sustained-first bonus

The dominance rate mirrors the Boot2Root acquisition/control shape:

$$D = 0.25L + 0.55S + 0.20\sqrt{LS}.$$

The local Leaderboard score is:

$$H^L = 100[Q + 0.50Q(1 - Q)D].$$

The first term pays for absolute challenge performance. The second rewards sustained first place. It contributes nothing when $Q=0$, cannot push perfect performance above 100, and is bounded by 12.5 points because $Q(1-Q) \leq 0.25$ and $D \leq 1$.

Table 3. Equal absolute performance with different first-place patterns.

Ten-tick pattern at $Q = 0.80$	L	S	D	Score
Never first	0.000	0.000	0.000	80.00
First on five alternating ticks	0.500	0.000	0.125	81.00
First for five consecutive ticks	0.500	0.444	0.464	83.71
First for all ten ticks	1.000	1.000	1.000	88.00

The alternating and consecutive teams reach first equally often, but only the second retains it across adjacent ticks. At $Q=0.20$ and $D=1$, the result is 28 rather than a winner jackpot:

$$100[0.20 + 0.50(0.20)(0.80)] = 28.$$

4.5 Same-tick rule

Consider two ticks. In tick one a team has activity 1.0 but objective performance 0 ; in tick two it has activity 0 and objective performance 1.0 . Both Q_t values are zero, so the epoch performance is zero. Computing the harmonic mean from the displayed epoch averages would incorrectly create a positive score. RSCTF therefore persists and averages the per-tick core.

5. HILL AGGREGATION, EPOCHS, AND RANK

Before scoring, each hill receives a frozen weight w_h in $[0.8, 1.2]$. Let $z_{he} = 1$ when hill h has field-wide scorable evidence in epoch e , and zero otherwise. The team epoch result is:

$$E_{ie} = \frac{\sum_h z_{he} w_h H_{ihe}}{\sum_h z_{he} w_h}.$$

A wholly void hill contributes no numerator or denominator. Once a Leaderboard hill has field evidence, an omitted team retains its explicit zero rather than removing the hill from its personal calculation.

A complete epoch has weight $q_e = 1$. If the event ends after r of n configured ticks in the final epoch, then $q_e = r/n$. The event score is:

$$T_i = \frac{\sum_e q_e E_{ie}}{\sum_e q_e}.$$

There is no late-epoch multiplier. Projected includes open epochs; Settled uses only finalized epochs.

Official KotH rank sorts by:

1. Settled score descending;
2. Control for Boot2Root or Objective rate for Leaderboard;
3. Reliability for Boot2Root or Sustained-lead rate for Leaderboard;
4. acquisition-window count or activity-positive-tick count; and
5. stable participation ID.

The live projection never breaks an official tie. Displayed KotH ranks are ordinal; the final ID provides deterministic ordering.

Table 4. Source-aware scoreboard metrics.

Badge	First metric	Second metric	Third metric	Crown state
Boot2Root	Acquisition	Control	Reliability	confirmed / pending
Leaderboard	Activity	Objective	Sustained lead	none

6. FAULT AND ADMISSION POLICY

Table 5. Evidence admission and fault treatment.

Condition	Boot2Root	Leaderboard
Missing own evidence in valid tick	no control/health credit	explicit zero row
Missing or changing whole snapshot	not applicable to marker feed	field-wide void
Functional checker fails globally	attributed only when responsibility is authoritative; otherwise void/recovery	field-wide void
Invalid/revoked capability	no claim	row rejected or unrecognized
Old cycle/reset/container	no claim	signed context rejected; event token remains usable after readiness
Replay or old timestamp	not a marker operation	request rejected
Objective IDs/order changed	not applicable	request rejected
Reset/readiness/incomplete issuance	void	void
Failed player exploit attempt	no score by itself	no negative points; telemetry/rate limits still apply

A **200** response from the signed endpoint only stages a snapshot. It does not award points. The checker still has to bracket the snapshot around a healthy probe and persist immutable evidence.

7. SECURITY AND ANTI-CHEAT CONTRACT

7.1 HMAC authenticates origin, not truth

Run the Leaderboard referee under a separate trusted identity and compute boundary. Its HMAC secret must not be present in the player image, environment, filesystem, browser, logs,

repository, or challenge backup. A valid signature shows that the referee produced an exact body; it does not show that the referee measured honestly.

Give the referee read-only access to the smallest evidence feed it needs, outbound access only to the arena and RSCTF, and persistent bounded cursor state. Prevent the player-facing arena from calling the signed endpoint. PostgreSQL backups contain the symmetric key and remain sensitive.

7.2 Make score require play

A defensible Leaderboard challenge:

1. counts completed challenge-relevant work, not requests, packets, page views, or connection volume;
2. issues unpredictable, expiring, one-use tasks or proofs;
3. binds each task and result to the capability hash that began it;
4. verifies results server-side and makes replay idempotent;
5. publishes fixed activity targets, objective IDs/order, and denominators;
6. bounds sessions, requests, evidence retention, pagination, and queue work;
7. keys team quotas by capability identity rather than source IP;
8. isolates per-team work so one team cannot exhaust shared admission;
9. exposes an ordered feed cursor and fails closed on a retention gap; and
10. keeps the functional checker independent and read-only.

Fuzzing, failed exploit development, and malformed challenge traffic are part of hacking unless event rules prohibit a specific action. They do not deserve an automatic score penalty. Rate limits and bounded work protect availability; only verified completed outcomes create positive scoring evidence.

7.3 Capability and context scope

RSCTF resolves a submitted `tokenHash` only against the current event capability for the exact game, hill, and official participation. The signed context independently binds the round, target, cycle, reset attempt, container, and frozen objective-schema hash. Raw bearer tokens enter only the narrowly scoped authentication exchange and never enter the signed referee request. Unknown and rotated hashes cannot become arbitrary team identities.

The wire path remains `/api/v1/...` for compatibility. `v1`, `Marker`, and `Api` are protocol or storage identifiers, not scoring-policy versions. The database stores no selectable KoH formula column.

8. ORGANIZER OPERATIONS

8.1 Preflight

Before official scoring:

- choose and publish each hill's format, cadence, bounded weight, checker contract, and appeal policy;

- confirm at least two accepted teams receive distinct current capabilities;
- for Boot2Root, run a pristine replacement and reject every previous cycle capability afterward;
- verify the checker before and after evidence reads;
- for Boot2Root, verify provisional confirmation, takeover, responsibility, and enforceable cooldown;
- for Leaderboard, verify the same event token works after a pristine reset, explicit rotation rejects the prior token and only that team's unsettled row, and then freeze the final objective IDs/order and verify HMAC, recognized/submitted equality, independent normalization, tied lead credit, explicit zero, referee restart, stale hash, replay, clock skew, feed gap, and changing-snapshot behavior; and
- run a complete accelerated epoch at expected peak capacity, including one deliberately overloaded team, while confirming that other teams and the referee retain reserved capacity.

The official snapshot records roster, hills, images, source, cadence, and weights. Objective identity freezes on the first accepted Leaderboard snapshot. These score-affecting inputs cannot change after the competition boundary.

8.2 During play

Monitor checker gaps, void reasons, reset phases, runtime identity, snapshot freshness, recognized-team counts, cursor lag, queue saturation, and referee errors. Pause scoring before rotating a referee secret, submit fresh evidence, verify it, then resume.

At event end, reject late evidence, complete authoritative in-flight checks within the deadline fence, settle the proportional tail, and publish awards from Settled results. Retain official configuration, capability issuance and revocation, immutable checker rows, Leaderboard tick evidence, lifecycle receipts, and epoch rollups through the appeal period.

8.3 Signed body summary

The complete wire contract and signing example are in the [organizer referee guide](#). The core body is:

```
{
  "context": "<64 lowercase hex characters>",
  "objectiveIds": ["proof-strength", "solve-speed"],
  "teams": [
    {
      "tokenHash": "<sha256 of current capability>",
      "activity": {"earned": 4, "possible": 5},
      "objectives": [
        {"earned": 7, "possible": 10},
        {"earned": 750, "possible": 1000}
      ]
    }
  ]
}
```

9. VALIDATION AND LIMITATIONS

The equations prove boundedness and zero conditions. Unit, database, and lifecycle tests establish behavior for exercised cases. Neither proves that an organizer chose competitively valid objectives.

Organizers should measure activity and objective distributions, lead concentration, missing/void frequency, referee lag, feed-retention margin, marker confirmation failures, control changes, repeat winners, reset latency, and score sensitivity to targets and cadence.

A randomized checker sample cannot prove continuous state between observations. Capability hashing cannot repair low-entropy tokens. HMAC cannot prove referee correctness. Equal normalization cannot prove equal objective difficulty. The formula cannot detect off-protocol collusion. A field-wide outage is void, so weak resource isolation could let one participant manufacture a veto. These remain challenge-design and operational responsibilities.

10. CONCLUSION

RSCTF does not treat concurrent KoH as Boot2Root with a different transport. Boot2Root rewards qualified acquisition, sustained exclusive control, and service reliability. Leaderboard rewards absolute normalized performance and adds a small, bounded premium for consistently remaining first.

Both formulas are constant and platform-owned. Dense zero rows prevent stale team scores; field-wide voids prevent infrastructure failures becoming participant penalties. Exact identity fences, independent checking, pristine replacement, bounded hill normalization, and finalized epochs make results reproducible. Fairness still depends on published rules, checker coverage, objective design, referee isolation, network equality, and incident review.

APPENDIX A. FREQUENTLY ASKED QUESTIONS

A.1 Is Leaderboard an observer for /koth/king ?

No. Boot2Root has one holder and acquisition/control/reliability evidence. Leaderboard has concurrent teams and activity/objective/sustained-lead evidence. Only the lifecycle infrastructure is shared.

A.2 Can the referee submit points or team IDs?

No. It submits integer evidence ratios and current capability hashes. RSCTF resolves identity, normalizes evidence, detects leaders, and computes points.

A.3 Why a harmonic mean?

It returns zero when either required play channel is zero and remains sensitive to a weak channel. The fixed 35/65 weights give objective performance greater influence without allowing it to replace activity.

A.4 Why normalize objectives separately?

Raw sums let the largest numeric unit dominate. Independent ratios give every objective the same **[0,1]** range before the equal mean.

A.5 What if my team is omitted from a valid snapshot?

RSCTF writes an explicit zero for that tick. It never repeats an old result.

A.6 What if the whole snapshot or application is unavailable?

The tick is field-wide void and enters no team's denominator.

A.7 Do failed hacking attempts lose points?

No. Only meaningful verified outcomes add evidence. Invalid traffic can be rate-limited and audited, but it is not an integrity score multiplier.

A.8 Does a reset erase settled evidence?

No. It removes transient containers, sessions, claims, and Boot2Root cycle capabilities. Leaderboard event tokens persist. Immutable results and finalized rollups remain.

A.9 Is there formula versioning?

No. Boot2Root and Leaderboard are distinct formats with one constant formula each. Protocol path **v1** is unrelated to scoring policy.

A.10 Can automation participate?

Yes, subject to event rules. RSCTF validates evidence and scope rather than guessing whether a participant is human or automated.

APPENDIX B. NOMENCLATURE

Table 6. KoH nomenclature.

Symbol or term	Definition
A, C, R	Boot2Root acquisition, control, and reliability rates
E_t, P_t	Leaderboard activity and normalized objective rates for tick t
Q_t, Q	per-tick harmonic performance and its epoch mean
l_t, L	tied-leader tick credit and epoch lead coverage
S, D	sustained-lead continuity and bounded dominance rate
H^M, H^L	local Boot2Root and Leaderboard scores in [0,100]
w_h	frozen hill weight in [0.8,1.2]

Symbol or term	Definition
z_{he}	field-evidence switch for one hill and epoch
q_e	complete or proportional final-epoch weight
Provisional	Boot2Root capability observed but not confirmed
Settled	official value using finalized epochs
Projected	information that also includes open evidence
Field void	sample excluded from every team's denominator
Explicit zero	omitted Leaderboard team in an otherwise valid tick

APPENDIX C. IMPLEMENTATION TRACEABILITY

Paths are relative to the repository revision containing this paper.

Table 7. Implementation traceability.

Responsibility	Source of truth
Official source, roster, hill, cadence, image, and weight snapshot	<code>src/services/ad/engine/koth_cycle/config.rs</code>
Durable pristine replacement	<code>src/services/ad/engine/koth_cycle/lifecycle/</code>
Boot2Root claim and acquisition	<code>src/services/ad/engine/koth_cycle/claims.rs</code>
Exact marker read	<code>src/services/ad/engine/koth_marker.rs</code>
Leaderboard event-token authentication and rotation	<code>src/services/ad/koth_api_capability.rs</code> , <code>src/controllers/game/koth/tokens.rs</code>
Signed context, HMAC, replay, schema, and submission	<code>src/controllers/game/koth/api/</code> , <code>api_contract.rs</code>
Stable snapshot read and tick core	<code>src/services/ad/engine/koth_api.rs</code>
Checker persistence, dense zeros, and tied leaders	<code>src/services/ad/engine/checker/koth_api.rs</code>
Constant pure formulas	<code>src/controllers/game/koth/scoring_formula.rs</code>
SQL epoch evidence and lead continuity	<code>src/controllers/game/koth/scoring/evidence.rs</code>
Final rollups	<code>src/controllers/game/koth/scoring/rollup/</code>
Board labels and rank	<code>src/controllers/game/koth/board.rs</code> , <code>web/src/components/KothScoreboardTable.tsx</code>
Constant Leaderboard schema and removal of formula selectors	<code>src/migrations/m0085_constant_leaderboard_scoring.rs</code>
Event-scoped Leaderboard token schema and live-token preservation	<code>src/migrations/m0086_koth_api_event_tokens.rs</code>

C.1 HTTP surface

Table 8. Player, operator, and referee HTTP surface.

Method and route	Purpose
GET <code>/api/game/{id}/ad/koth/{challengeId}/token</code>	caller's exact current-hill capability

Method and route	Purpose
POST <code>/api/game/{id}/ad/koth/{challengeId}/token</code>	explicitly rotate a Leaderboard event capability
POST <code>/api/v1/koth/capability/authenticate</code>	exchange one scoped event capability for the authoritative arena identity
GET <code>/api/game/{id}/ad/koth/{challengeId}/state</code>	lifecycle and Boot2Root holder state
GET <code>/api/game/{id}/ad/koth/scoreboard</code>	source-aware metrics, Projected, Settled, rank
GET <code>/api/game/{id}/ad/koth/timeline</code>	finalized/projected cumulative history
GET <code>/api/edit/games/{id}/ad/koth/state</code>	operator lifecycle and evidence view
POST <code>/api/edit/games/{id}/ad/koth/{challengeId}/recover</code>	idempotent lifecycle recovery
GET/POST/DELETE <code>/api/edit/games/{id}/ad/koth/{challengeId}/observer</code>	inspect, create/rotate once, or revoke credential
GET <code>/api/v1/koth/games/{id}/challenges/{challengeId}/context</code>	exact runtime/schema fence and eligible hashes
POST <code>/api/v1/koth/games/{id}/challenges/{challengeId}/observations</code>	signed bounded evidence; never points

Wire DTOs use camelCase and Unix-millisecond timestamps.

C.2 Verification scope

The suite covers formula bounds and zero conditions, objective normalization, ordered schema identity, malformed evidence, HMAC scope, clock skew, replay, context rotation, current capability resolution, unknown hashes, dense zeros, snapshot bracketing, tied lead credit, continuity, marker confirmation, cooldown denominators, tied champions, partial epochs, rollups, and ordinal ranks. The bundled challenge exercises one-use proofs, invalid traffic, capability hashing, bounded pagination, persistent referee restart, feed-gap failure, and exact signed bodies. Lifecycle and fixed-load results are recorded in `tests/load/REPORT.md`; they do not establish competitive fairness.

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